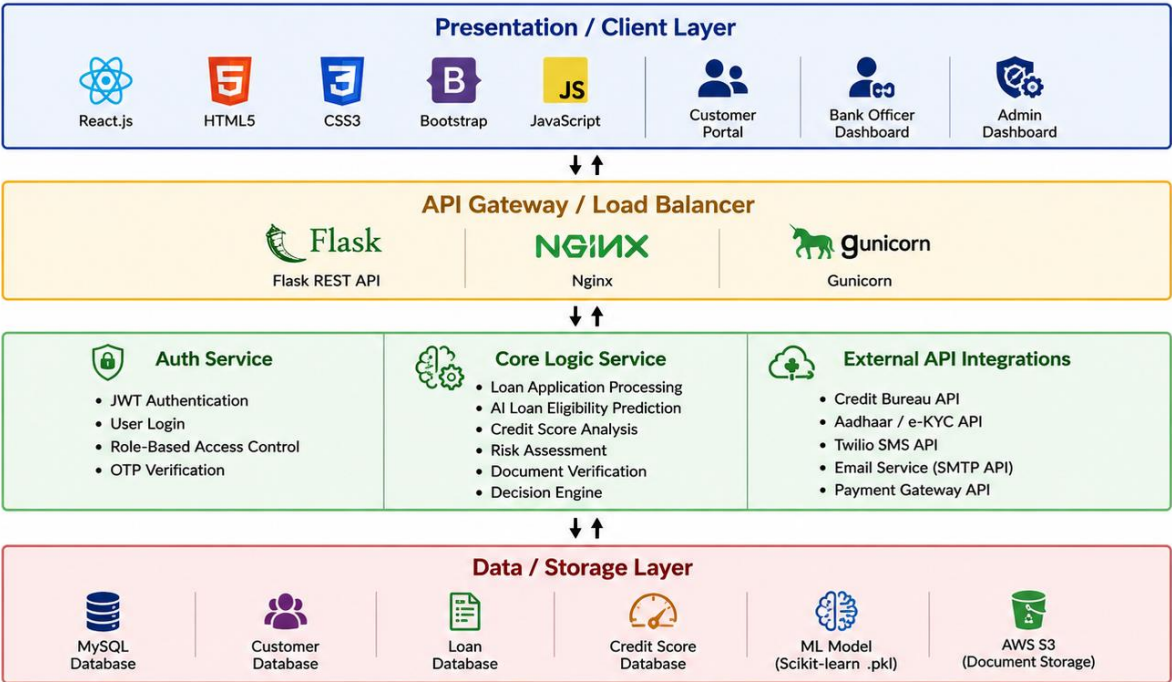


Solution Architecture

Date	15 March 2026
Team ID	
Project Name	Smart Lender: AI/ML-Based Loan Approval Prediction System

Solution Architecture Diagram:



Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for customers, bank officers, and administrators to apply for loans, track applications, and manage the system.	React.js, HTML5, CSS3, Bootstrap, JavaScript
API Gateway	Acts as a single entry point for all client requests, routes them to appropriate services, and handles load balancing.	Flask REST API, Nginx, Gunicorn
Microservices	Handles authentication, business logic, AI prediction, document verification, risk assessment, and integration with external APIs.	Python, Flask, Scikit-learn, Pandas, NumPy, JWT
Database	Stores all application data including customer details, loan applications, credit data, and system logs.	MySQL, AWS S3 (Optional)

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides user interface for customers, bank officers, and administrators.	React.js, HTML5, CSS3, Bootstrap, JavaScript
API Gateway	Routes client requests to backend services and manages APIs securely.	Flask REST API, Gunicorn, Nginx

Microservices	Handles authentication, AI loan prediction, business logic, document verification, and external API communication.	Python, Flask, Scikit-learn, Pandas, NumPy, JWT
Database	Stores customer records, loan applications, AI predictions, and transaction data.	MySQL, AWS S3 (Optional)